

```
In [2]: import numpy as np
```

```
In [4]: np.__version__
```

```
Out[4]: '2.4.2'
```

Creating List

```
In [5]: my_list = [0,1,2,3,4,5]
my_list
```

```
Out[5]: [0, 1, 2, 3, 4, 5]
```

```
In [6]: type(my_list)
```

```
Out[6]: list
```

```
In [7]: # how to convert list to array
arr = np.array(my_list)
arr
```

```
Out[7]: array([0, 1, 2, 3, 4, 5])
```

```
In [8]: type(arr)
```

```
Out[8]: numpy.ndarray
```

```
In [9]: print(type(my_list))
print(type(arr))
```

```
<class 'list'>
<class 'numpy.ndarray'>
```

```
In [11]: np.arange(5)
```

```
Out[11]: array([0, 1, 2, 3, 4])
```

```
In [12]: np.arange(5,10)
```

```
Out[12]: array([5, 6, 7, 8, 9])
```

```
In [13]: np.arange(5,100,10)
```

```
Out[13]: array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

```
In [14]: np.arange(20,10)
```

```
Out[14]: array([], dtype=int64)
```

```
In [15]: np.arange(-20,10)
```

```
Out[15]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
               -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
                6,  7,  8,  9])
```

```
In [4]: np.arange(10,100,10)
```

```
Out[4]: array([10, 20, 30, 40, 50, 60, 70, 80, 90])
```

```
In [3]: np.arange(10)
```

```
Out[3]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [6]: np.zeros(5)
```

```
Out[6]: array([0., 0., 0., 0., 0.])
```

```
In [7]: np.zeros(5, dtype = int)
```

```
Out[7]: array([0, 0, 0, 0, 0])
```

```
In [8]: np.zeros((2,2))
```

```
Out[8]: array([[0., 0.],  
               [0., 0.]])
```

```
In [9]: np.zeros((2,3),dtype = int)
```

```
Out[9]: array([[0, 0, 0],  
               [0, 0, 0]])
```

```
In [10]: np.zeros((10,3),dtype = int)
```

```
Out[10]: array([[0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0],  
                [0, 0, 0]])
```

```
In [11]: np.zeros((10,10),dtype = int)
```

```
Out[11]: array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
                [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

```
In [4]: np.ones(4)
```

```
Out[4]: array([1., 1., 1., 1.])
```

```
In [5]: np.ones((2,3), dtype = int)
```

```
Out[5]: array([[1, 1, 1],
               [1, 1, 1]])
```

```
In [6]: np.two((2,3))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[6], line 1
----> 1 np.two((2,3))

File ~\AppData\Roaming\Python\Python314\site-packages\numpy\__init__.py:805, in _
_getattr__(attr)
      802     import numpy.char as char
      803     return char.chararray
--> 805 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'two'
```

```
In [7]: np.three((2,3))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[7], line 1
----> 1 np.three((2,3))

File ~\AppData\Roaming\Python\Python314\site-packages\numpy\__init__.py:805, in _
_getattr__(attr)
      802     import numpy.char as char
      803     return char.chararray
--> 805 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'three'
```

```
In [8]: np.four((2,3))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[8], line 1
----> 1 np.four((2,3))

File ~\AppData\Roaming\Python\Python314\site-packages\numpy\__init__.py:805, in _
_getattr__(attr)
      802     import numpy.char as char
      803     return char.chararray
--> 805 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'four'
```

generating OTP

```
In [9]: rand(2,3)
```

```
-----
NameError                                    Traceback (most recent call last)
Cell In[9], line 1
----> 1 rand(2,3)

NameError: name 'rand' is not defined
```

```
In [10]: random.rand(2,3)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[10], line 1
----> 1 random.rand(2,3)

NameError: name 'random' is not defined
```

```
In [12]: np.random.rand(2)
```

```
Out[12]: array([0.1555956 , 0.62618822])
```

```
In [15]: np.random.rand(2,3)
```

```
Out[15]: array([[0.69039738, 0.24288155, 0.76018878],
               [0.45110856, 0.84052392, 0.56165421]])
```

```
In [16]: np.random.randint(2)
```

```
Out[16]: 0
```

```
In [21]: np.random.randint(2,9)
```

```
Out[21]: 2
```

```
In [27]: import numpy as np
def generate_otp(length = 6):
    #generate otp of given length using np.random.randint
    otp = ''.join([str(np.random.randint(0,10)) for _ in range(length)])
    return otp
    #Example usage
print("Your OTP is:", generate_otp())
```

Your OTP is: 017043

```
In [28]: import numpy as np
def generate_otp(length = 4):
    #generate otp of given length using np.random.randint
    otp = ''.join([str(np.random.randint(0,10)) for _ in range(length)])
    return otp
    #Example usage
print("Your OTP is:", generate_otp())
```

Your OTP is: 3739

```
In [36]: np.random.randint(10,40,(10,10)) # 10*10 matrix between 10 to 40
```

```
Out[36]: array([[17, 21, 35, 21, 18, 23, 20, 27, 35, 35],
               [16, 39, 10, 24, 11, 37, 33, 37, 27, 30],
               [27, 15, 27, 22, 35, 27, 28, 21, 31, 35],
               [19, 18, 27, 29, 21, 16, 35, 17, 26, 36],
               [22, 12, 23, 34, 23, 10, 30, 35, 12, 19],
               [14, 13, 19, 34, 31, 11, 33, 30, 22, 21],
               [32, 21, 36, 27, 28, 35, 37, 14, 24, 34],
               [32, 39, 12, 28, 14, 11, 32, 23, 12, 14],
               [18, 39, 20, 30, 12, 31, 32, 26, 38, 16],
               [31, 16, 38, 33, 37, 30, 32, 24, 33, 12]], dtype=int32)
```

```
In [39]: arr1 = np.random.randint(10,40,(10,10))
print(arr1)
```

```
[[14 23 14 28 15 11 14 31 11 20]
 [28 35 36 25 24 36 34 32 37 37]
 [17 35 14 10 36 37 15 34 22 26]
 [29 31 12 36 36 20 14 32 39 13]
 [27 16 15 29 39 22 21 29 21 18]
 [10 35 13 13 25 16 38 25 37 37]
 [12 39 35 10 29 23 34 38 11 32]
 [13 19 37 31 22 24 16 36 18 24]
 [31 28 27 28 20 28 38 17 12 20]
 [24 36 18 37 38 34 13 25 31 11]]
```

```
In [40]: print(arr1)
```

```
[[14 23 14 28 15 11 14 31 11 20]
 [28 35 36 25 24 36 34 32 37 37]
 [17 35 14 10 36 37 15 34 22 26]
 [29 31 12 36 36 20 14 32 39 13]
 [27 16 15 29 39 22 21 29 21 18]
 [10 35 13 13 25 16 38 25 37 37]
 [12 39 35 10 29 23 34 38 11 32]
 [13 19 37 31 22 24 16 36 18 24]
 [31 28 27 28 20 28 38 17 12 20]
 [24 36 18 37 38 34 13 25 31 11]]
```

```
In [42]: arr1[5]
```

```
Out[42]: array([10, 35, 13, 13, 25, 16, 38, 25, 37, 37], dtype=int32)
```

```
In [44]: arr2 = np.random.randint(2,9,(5,7))
print(arr2)
```

```
[[4 4 3 4 3 7 6]
 [5 4 8 2 2 3 6]
 [8 4 4 7 6 6 2]
 [4 7 8 8 6 6 7]
 [3 3 8 7 5 5 7]]
```

```
In [45]: arr2[-3]
```

```
Out[45]: array([8, 4, 4, 7, 6, 6, 2], dtype=int32)
```

```
In [46]: arr2
```

```
Out[46]: array([[4, 4, 3, 4, 3, 7, 6],
                [5, 4, 8, 2, 2, 3, 6],
                [8, 4, 4, 7, 6, 6, 2],
                [4, 7, 8, 8, 6, 6, 7],
                [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [47]: arr2[2:10]
```

```
Out[47]: array([[8, 4, 4, 7, 6, 6, 2],
                [4, 7, 8, 8, 6, 6, 7],
                [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [48]: arr2
```

```
Out[48]: array([[4, 4, 3, 4, 3, 7, 6],
               [5, 4, 8, 2, 2, 3, 6],
               [8, 4, 4, 7, 6, 6, 2],
               [4, 7, 8, 8, 6, 6, 7],
               [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [49]: arr2[:]
```

```
Out[49]: array([[4, 4, 3, 4, 3, 7, 6],
               [5, 4, 8, 2, 2, 3, 6],
               [8, 4, 4, 7, 6, 6, 2],
               [4, 7, 8, 8, 6, 6, 7],
               [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [50]: arr2
```

```
Out[50]: array([[4, 4, 3, 4, 3, 7, 6],
               [5, 4, 8, 2, 2, 3, 6],
               [8, 4, 4, 7, 6, 6, 2],
               [4, 7, 8, 8, 6, 6, 7],
               [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [51]: arr2[::-1]
```

```
Out[51]: array([[3, 3, 8, 7, 5, 5, 7],
               [4, 7, 8, 8, 6, 6, 7],
               [8, 4, 4, 7, 6, 6, 2],
               [5, 4, 8, 2, 2, 3, 6],
               [4, 4, 3, 4, 3, 7, 6]], dtype=int32)
```

```
In [52]: arr2
```

```
Out[52]: array([[4, 4, 3, 4, 3, 7, 6],
               [5, 4, 8, 2, 2, 3, 6],
               [8, 4, 4, 7, 6, 6, 2],
               [4, 7, 8, 8, 6, 6, 7],
               [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [53]: arr2[:,3]
```

```
Out[53]: array([[4, 4, 3, 4, 3, 7, 6],
               [4, 7, 8, 8, 6, 6, 7]], dtype=int32)
```

```
In [54]: arr2
```

```
Out[54]: array([[4, 4, 3, 4, 3, 7, 6],
               [5, 4, 8, 2, 2, 3, 6],
               [8, 4, 4, 7, 6, 6, 2],
               [4, 7, 8, 8, 6, 6, 7],
               [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [55]: arr2
```

```
Out[55]: array([[4, 4, 3, 4, 3, 7, 6],
               [5, 4, 8, 2, 2, 3, 6],
               [8, 4, 4, 7, 6, 6, 2],
               [4, 7, 8, 8, 6, 6, 7],
               [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [56]: arr2[3,4]
```

```
Out[56]: np.int32(6)
```

```
In [57]: print(arr2[3,4])
```

6

```
In [58]: arr2
```

```
Out[58]: array([[4, 4, 3, 4, 3, 7, 6],
                [5, 4, 8, 2, 2, 3, 6],
                [8, 4, 4, 7, 6, 6, 2],
                [4, 7, 8, 8, 6, 6, 7],
                [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [59]: print(arr2[-4,-5])
```

8

```
In [60]: arr2
```

```
Out[60]: array([[4, 4, 3, 4, 3, 7, 6],
                [5, 4, 8, 2, 2, 3, 6],
                [8, 4, 4, 7, 6, 6, 2],
                [4, 7, 8, 8, 6, 6, 7],
                [3, 3, 8, 7, 5, 5, 7]], dtype=int32)
```

```
In [63]: arr2[0,1] = 10
```

```
In [64]: arr2
```

```
Out[64]: array([[ 4, 10, 10,  4,  3,  7,  6],
                [ 5,  4,  8,  2,  2,  3,  6],
                [ 8,  4,  4,  7,  6,  6,  2],
                [ 4,  7,  8,  8,  6,  6,  7],
                [ 3,  3,  8,  7,  5,  5,  7]], dtype=int32)
```

```
In [65]: arr2[:,0]
```

```
Out[65]: array([4, 5, 8, 4, 3], dtype=int32)
```

```
In [66]: arr2[:, -3]
```

```
Out[66]: array([3, 2, 6, 6, 5], dtype=int32)
```

```
In [67]: arr2
```

```
Out[67]: array([[ 4, 10, 10,  4,  3,  7,  6],
                [ 5,  4,  8,  2,  2,  3,  6],
                [ 8,  4,  4,  7,  6,  6,  2],
                [ 4,  7,  8,  8,  6,  6,  7],
                [ 3,  3,  8,  7,  5,  5,  7]], dtype=int32)
```

```
In [69]: arr2[2:5,2:5]
```

```
Out[69]: array([[4, 7, 6],
                [8, 8, 6],
                [8, 7, 5]], dtype=int32)
```

```
In [70]: arr2
```

```
Out[70]: array([[ 4, 10, 10,  4,  3,  7,  6],
               [ 5,  4,  8,  2,  2,  3,  6],
               [ 8,  4,  4,  7,  6,  6,  2],
               [ 4,  7,  8,  8,  6,  6,  7],
               [ 3,  3,  8,  7,  5,  5,  7]], dtype=int32)
```

```
In [71]: arr2[1:4,2:4]
```

```
Out[71]: array([[8, 2],
               [4, 7],
               [8, 8]], dtype=int32)
```

```
In [72]: # masking or filter in numpy
arr2
```

```
Out[72]: array([[ 4, 10, 10,  4,  3,  7,  6],
               [ 5,  4,  8,  2,  2,  3,  6],
               [ 8,  4,  4,  7,  6,  6,  2],
               [ 4,  7,  8,  8,  6,  6,  7],
               [ 3,  3,  8,  7,  5,  5,  7]], dtype=int32)
```

```
In [73]: arr2 > 5
```

```
Out[73]: array([[False,  True,  True, False, False,  True,  True],
               [False, False,  True, False, False, False,  True],
               [ True, False, False,  True,  True,  True, False],
               [False,  True,  True,  True,  True,  True,  True],
               [False, False,  True,  True, False, False,  True]])
```

```
In [74]: arr2[arr2 > 5]
```

```
Out[74]: array([10, 10,  7,  6,  8,  6,  8,  7,  6,  6,  7,  8,  8,  6,  6,  7,  8,
               7,  7], dtype=int32)
```

```
In [75]: arr2[arr2 >= 5]
```

```
Out[75]: array([10, 10,  7,  6,  5,  8,  6,  8,  7,  6,  6,  7,  8,  8,  6,  6,  7,
               8,  7,  5,  5,  7], dtype=int32)
```

```
In [76]: arr2
```

```
Out[76]: array([[ 4, 10, 10,  4,  3,  7,  6],
               [ 5,  4,  8,  2,  2,  3,  6],
               [ 8,  4,  4,  7,  6,  6,  2],
               [ 4,  7,  8,  8,  6,  6,  7],
               [ 3,  3,  8,  7,  5,  5,  7]], dtype=int32)
```

```
In [77]: arr2[arr2 == 5]
```

```
Out[77]: array([5, 5, 5], dtype=int32)
```

```
In [79]: arr2[arr2 != 5]
```

```
Out[79]: array([ 4, 10, 10,  4,  3,  7,  6,  4,  8,  2,  2,  3,  6,  8,  4,  4,  7,
               6,  6,  2,  4,  7,  8,  8,  6,  6,  7,  3,  3,  8,  7,  7],
               dtype=int32)
```

```
In [82]: arr2
```

```
Out[82]: array([[ 4, 10, 10,  4,  3,  7,  6],
                [ 5,  4,  8,  2,  2,  3,  6],
                [ 8,  4,  4,  7,  6,  6,  2],
                [ 4,  7,  8,  8,  6,  6,  7],
                [ 3,  3,  8,  7,  5,  5,  7]], dtype=int32)
```

```
In [87]: arr
```

```
Out[87]: array([0, 1, 2, 3, 4, 5])
```

```
In [88]: arr.reshape(2,3)
```

```
Out[88]: array([[0, 1, 2],
                [3, 4, 5]])
```

```
In [89]: arr.reshape(4,3)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[89], line 1
----> 1 arr.reshape(4,3)

ValueError: cannot reshape array of size 6 into shape (4,3)
```

```
In [90]: np.reshape(10,1)
```

```
Out[90]: array([10])
```

```
In [91]: np.reshape(6,1)
```

```
Out[91]: array([6])
```

```
In [92]: arr.reshape(6,1)
```

```
Out[92]: array([[0],
                [1],
                [2],
                [3],
                [4],
                [5]])
```

```
In [93]: arr.reshape(1,6)
```

```
Out[93]: array([[0, 1, 2, 3, 4, 5]])
```

```
In [94]: arr.reshape(3,2)
```

```
Out[94]: array([[0, 1],
                [2, 3],
                [4, 5]])
```

```
In [95]: arr.max()
```

```
Out[95]: np.int64(5)
```

```
In [96]: arr.min()
```

```
Out[96]: np.int64(0)
```

